Problem 8

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Problem 1 *Suppose that a stone is thrown vertically upward from a cliff on Mars with an initial velocity of 24 ft/s from a height of 192 ft. The height s of the stone above the ground after t seconds is given by $s(t) = -6t^2 + 24t + 192$.*

(a) *Determine the velocity and acceleration of the stone after t seconds.*

Explanation. The velocity $v(t)$ is: $v(t) = s'(t) = -12t + 24$.

The acceleration $a(t)$ is: $a(t) = v'(t) = s''(t) = -12$.

(b) *What is the greatest height of the stone and when does it occur? What are the velocity and acceleration at that time?*

Explanation. Since the function $s(t)$ is differentiable everywhere (it is a polynomial), the maximum height must occur at a time when the velocity is 0. So we solve:

$$\begin{aligned} v(t) = -12t + 24 &= 0 \\ 12t &= 24 \\ t &= 2 \end{aligned}$$

It is easy to check that $v(t) > 0$ for $0 \leq t < 2$ and $v(t) < 0$ for $2 < t$, and so the greatest height of the stone really does occur at time $t = 2$. The greatest height is $s(2) = -24 + 48 + 192 = 216$ ft. For the second question we have already seen that $v(2) = 0$ ft/sec, and we have a constant acceleration of -12 ft/sec^2. So $a(2) = -12$ ft/sec^2.

(c) *When does the stone hit the ground? What are the velocity and acceleration at that time?*

Explanation. The stone hits the ground when $s(t) = 0$. So we solve:

$$\begin{aligned} s(t) = -6t^2 + 24t + 192 &= 0 \\ -6(t^2 - 4t - 32) &= 0 \\ -6(t-8)(t+4) &= 0 \end{aligned}$$

Since we are only considering $t \geq 0$, we must have $t = 8$. At this instant, $v(8) = -96 + 24 = -72$ ft/sec and $a(8) = -12$ ft/sec^2.